
Stochastic economies and estimation

Adrien Auclert

Goethe Heterogeneous-Agent Macro Workshop, 2026

Introduction

- ❖ **Workshop so far:** MIT shocks (nonlinear and linear)
- ❖ Fine object for very rare events, but for business-cycle-like shocks, **rational expectations** is more useful benchmark
 - ❖ How useful are MIT shocks for rational expectations solution?
- ❖ Our answer (over today and tomorrow's lectures): **very useful!**
 - ❖ In fact, for practical purposes, **MIT shocks are all you need!** ("certainty correspondence")
 - ❖ Today: first instance of this: MIT shocks are all you need **to first order** ("certainty equivalence"). Apply this to **simulations + estimation.**

First-order certainty equivalence

Setup

- ❖ Consider general macro model:

$$F\left(X_{t-1}, X_t, \mathbb{E}_t[X_{t+1}], \epsilon_t\right) = 0$$

- ❖ $X_t \in \mathbb{R}^n$ are endogenous variables [het-agent applications: n large, eg $n = 2,000$]
- ❖ $\epsilon_t \in \mathbb{R}$ are exogenous innovations (realized shock process) [naturally extends to >1 shocks]
 - ❖ Distributed $\epsilon_t = \sigma_R \cdot \bar{\epsilon}_t$ with $\bar{\epsilon}_t$ iid, mean 0, variance 1, and symmetric density
- ❖ Expectations $\mathbb{E}_t[\cdot]$ are taken assuming future distribution of ϵ_t is $\sigma \cdot \bar{\epsilon}_t$
 - ❖ **Rational expectations:** $\sigma = \sigma_R$ (perceived distribution of future $\epsilon_t = \text{truth}$)
 - ❖ **Perfect foresight:** $\sigma = 0$ (agents think all future ϵ_t will realize to 0 for sure)
- ❖ **Deterministic steady state** $X \in \mathbb{R}^n$ solves $F(X, X, X, 0) = 0$ (i.e. $\sigma = \sigma_R = 0$)

PE SIM example

- ❖ For concreteness, consider the PE SIM model

$$V_t(e, a) = \max_{c, a'} u(c) + \beta \mathbb{E}_{e'}[V_{t+1}(e', a') | e]$$

$$\text{s.t.} \quad a' + c = (1 + r)a + Z_t e \quad a' \geq 0$$

- ❖ Aggregate income Z_t follows AR(1) process: $Z_t - 1 = \rho(Z_{t-1} - 1) + \epsilon_t$
- ❖ Here, $X_t \equiv (\mathbf{v}_t, \boldsymbol{\phi}_t, C_t, Z_t)$ where $\mathbf{v}_t$ stores marginal value $V_{a,t}$ over gridpoints (e, a) , $\boldsymbol{\phi}_t$ stores the p.d.f, and C_t aggregates consumption policy over distribution
- ❖ Equations describing F : the law of motion of Z_t , plus

Dynamic programming:
expectations enter via $\mathbb{E}_t[\mathbf{v}_{t+1}]$

$$\mathbf{v}_t = \mathcal{V}(Z_t, \mathbb{E}_t \mathbf{v}_{t+1})$$

$$\boldsymbol{\phi}_t = \Lambda(Z_t, \mathbb{E}_t \mathbf{v}_{t+1}, \boldsymbol{\phi}_{t-1})$$

$$C_t = \mathcal{C}(Z_t, \mathbb{E}_t \mathbf{v}_{t+1}, \boldsymbol{\phi}_{t-1})$$

(Stochastic) sequence space solution

- ❖ Given σ , consider **nonlinear sequence-space solution** to F :

$$X_t = \mathcal{X} \left(\sigma; \epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, \dots \right) \quad [\text{Lan-Meyer-Gohde, Lombardo-Uhlig}]$$

[eg, obtained from stable state-space solution $X_t = g \left(X_{t-1}, \epsilon_t; \sigma \right)$]

- ❖ We are going to relate this to the **perfect-foresight solution**

$$X_t^{PF} = \mathcal{X} \left(0; \epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, \dots \right)$$

[eg, obtained from ∞ -long sequence of repeated-surprise MIT shocks]

- ❖ Use **perturbation approach**: linearize $\mathcal{X}$ around $\sigma = \sigma_R = 0$ (i.e. $\epsilon = \mathbf{0}$)

- ❖ Helpful result: $\mathcal{X} \left(\sigma; \epsilon_t, \epsilon_{t-1}, \dots \right) = \mathcal{X} \left(-\sigma; \epsilon_t, \epsilon_{t-1}, \dots \right)$, so, e.g., $\mathcal{X}_\sigma(\mathbf{0}) = 0$

MIT shock impulse response

- ❖ Special case of X^{PF} is the MIT shock impulse response

$$X_j^{MIT}(\nu) \equiv \mathcal{X}(0; 0, \dots, 0, \underset{\substack{\uparrow \\ \text{position } j}}{\nu}, 0, \dots)$$

- ❖ Can compute $\{X_j^{MIT}(\nu)\}_j$ with nonlinear perfect foresight starting from s.s.
 - ❖ cf lecture 1 and 3 methods!
- ❖ **Next:** why this is sufficient to get the first-order solution with aggregate risk!

Certainty equivalence

- ❖ Expand $X_t = \mathcal{X}(\sigma, \epsilon_t, \epsilon_{t-1}, \dots)$ to first order in (σ, ϵ) : No anticipated risk,
no past realization

$$X_t = X + \cancel{\mathcal{X}_\sigma \sigma} + \frac{\partial \mathcal{X}}{\partial \epsilon_0}(\mathbf{0}) \epsilon_t + \frac{\partial \mathcal{X}}{\partial \epsilon_{-1}}(\mathbf{0}) \epsilon_{t-1} + \dots + \mathcal{O}(\sigma^2)$$

Symmetry:
Steady state unchanged

- ❖ So, to first order, X_t follows **linear MA** process, with coefficients equal to

$$\frac{\partial \mathcal{X}}{\partial \epsilon_{-j}}(\mathbf{0}) = \frac{dX_j^{MIT}}{d\nu}(0) \equiv \mathcal{X}_j$$

Certainty equivalence. To first order, X_t follows a linear MA process whose coefficients are given by the first order MIT shock impulse response of X to ϵ [Boppart-Krusell-Mitman]

Implementation using SSJ

- ❖ To first-order, X_t (so all its elements!) follows linear $MA(\infty)$

$$X_t = X + \mathcal{X}_0 \epsilon_t + \mathcal{X}_1 \epsilon_{t-1} + \mathcal{X}_2 \epsilon_{t-2} + \cdots + \mathcal{O}(\sigma^2) \quad \mathcal{X}_j \equiv \frac{dX_j^{MIT}}{d\nu}(0)$$

- ❖ Convenient: we can use SSJ to very effectively get $\mathcal{X}_j$!
- ❖ For instance, suppose that we want the behavior of aggregate outcome X_t to monetary shock r_s in the canonical HANK model, with $MA(\infty)$ shock process

$$r_t = r + \mathcal{R}_0 \epsilon_t + \mathcal{R}_1 \epsilon_{t-1} + \mathcal{R}_2 \epsilon_{t-2} + \cdots +$$

- ❖ Then we get the SSJ $G^{X,r}$ of X to r , then apply vector-matrix multiplication

$$(\mathcal{X}_0, \mathcal{X}_1, \mathcal{X}_2, \cdots)' = G^{X,r} \cdot (\mathcal{R}_0, \mathcal{R}_1, \mathcal{R}_2, \cdots)'$$

Applications: simulation, second moments,
estimation

Application to simulation

- ❖ Suppose we want to do stochastic simulations of X_t . Since

$$X_t \simeq X + \mathcal{X}_0 \epsilon_t + \mathcal{X}_1 \epsilon_{t-1} + \mathcal{X}_2 \epsilon_{t-2} + \dots \quad (\star)$$

- ❖ Once we have $\mathcal{X}_t$, we can just simulate by applying $(\star)!$

```
rho = 0.9
dishock = rho*np.arange(T)
dY = G['Y', 'ishock'] @ dishock
```

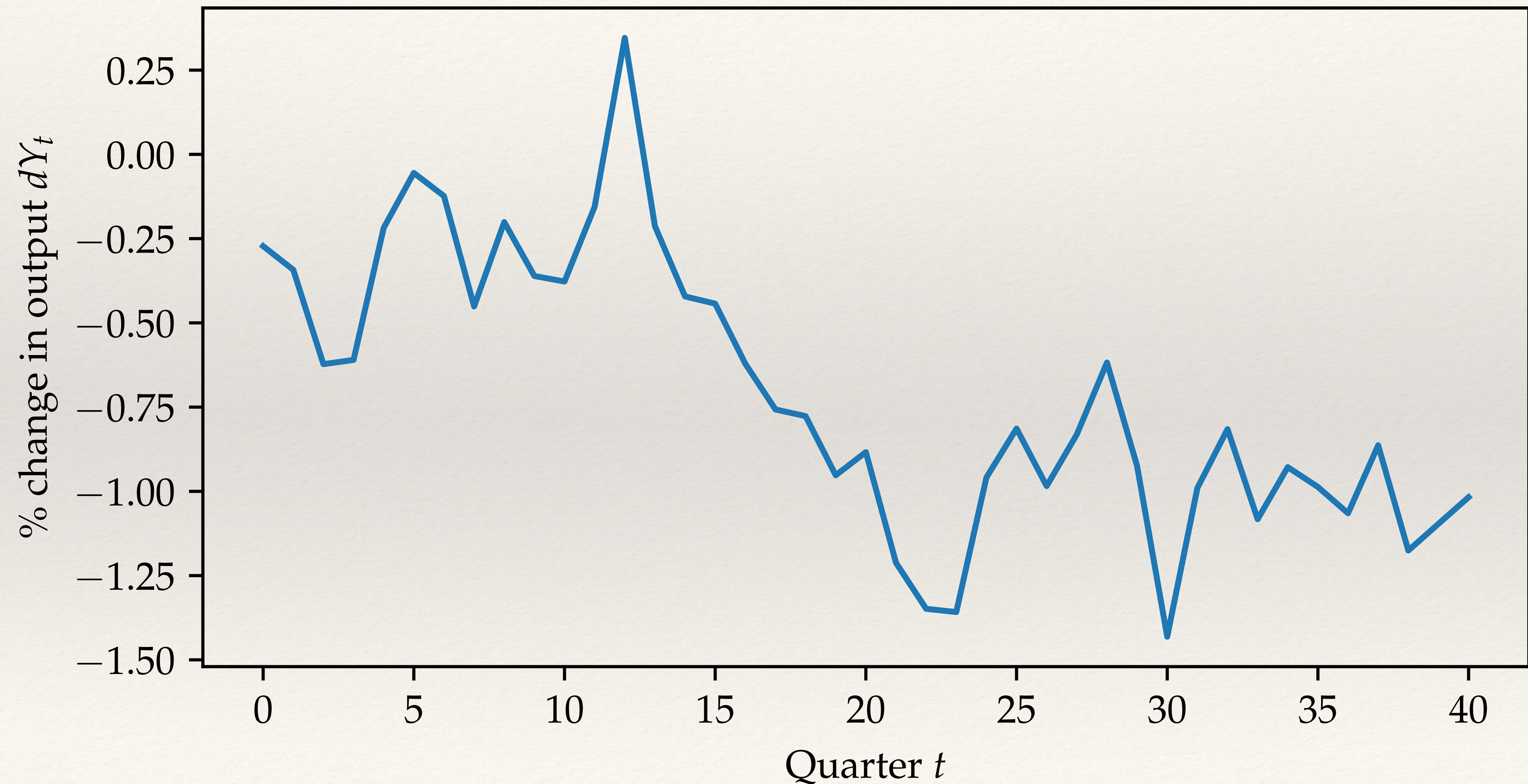
✓ 0.0s

```
sigma = 0.005
Tsim = 1_000_000 # will simulate for 1 million quarters
np.random.seed(40)
eps = np.random.randn(Tsim+T) # draw epsilons for T additional pre-per
dY_sim = np.empty(Tsim)
for t in range(Tsim):
    # at each t, take window of current up to T-1 previous epsilons
    # take dot product with MA coefficients implied by MIT shock
    dY_sim[t] = np.dot(dY, eps[t:t+T][::-1]) * sigma
```

✓ 0.7s

Python

Simulating output response to monetary shock



Application to second moments

- ❖ Suppose we want to obtain second moments of X_t (with $\sigma_R = \sigma$). Have

$$X_t \simeq X + \mathcal{X}_0 \epsilon_t + \mathcal{X}_1 \epsilon_{t-1} + \mathcal{X}_2 \epsilon_{t-2} + \cdots \quad (\star)$$

and we can just apply time series formulas to get these!

- ❖ For instance ergodic variance is area under squared impulse response

$$\text{Var}(X_t) = \sigma^2 \sum_{j=0}^{\infty} \left(\mathcal{X}_j \right)^2$$

- ❖ More generally

$$\text{Cov}(X_t, X_{t-k}) = \sigma^2 \sum_{j=0}^{\infty} \mathcal{X}_j \mathcal{X}_{j+k}$$

Implementation: second moments in HANK model

```
var_dY = sigma**2 * (dY @ dY)
```

Python

Compare to the sample variance to make sure this is right. Note that it's close but not exact, reflecting Monte Carlo error:

```
sample_var_dY = (dY_sim - dY_sim.mean()) @ (dY_sim - dY_sim.mean()) / Tsim  
print(f"Sample variance: {sample_var_dY:.4e} | theoretical : {var_dY:.4e}")
```

Python

```
Sample variance: 3.1761e-05 | theoretical : 3.1870e-05
```

Application to impulse-response matching

- ❖ Straightforward to estimate by **matching impulse responses**
- ❖ Suppose model parameters are θ , giving impulse response $IRF_t(\theta)$, and identified empirical impulse response $\hat{irf}_t$

- ❖ Then we can just search for

$$\hat{\theta} = \arg \min \left(IRF(\theta) - \hat{irf} \right)' V^{-1} \left(IRF(\theta) - \hat{irf} \right)$$

where, e.g. V has sample variances of $\hat{irf}_t$ (Christiano, Eichenbaum, Evans)

- ❖ This is especially fast if we do not have to recompute all Jacobians for $IRF(\theta)$, such as we estimate shock process parameters (G unchanged!)

Application to method of moments

- ❖ Also straightforward to estimate by **matching moments**
- ❖ Suppose model parameters are θ , giving model moments $M(\theta)$
 - ❖ eg second moments, HP-filtered moments, regression coefficients...
- ❖ Then we can confront to empirical counterparts $\hat{m}$ by taking

$$\hat{\theta} = \arg \min (M(\theta) - \hat{m})' V^{-1} (M(\theta) - \hat{m})'$$

where, e.g. V is model's $\mathbb{E} [M(\theta) M(\theta)']$

(Minimum distance estimation, aka “simulated” method of moments)

Application to likelihood-based estimation

- ❖ Let's assume that model innovations ϵ_t are normally distributed
- ❖ Given parameter θ , first-order process followed by X_t is

$$X_t \simeq X(\theta) + \mathcal{X}_0(\theta)\epsilon_t + \mathcal{X}_1(\theta)\epsilon_{t-1} + \mathcal{X}_2(\theta)\epsilon_{t-2} + \dots$$

- ❖ Given data $\mathbf{X}$, we know the **(log) likelihood function**

$$\mathcal{L}(\mathbf{X}; \theta) = -\frac{1}{2} \log \det \mathbf{V}(\theta) - \frac{1}{2} \mathbf{X}' \mathbf{V}(\theta)^{-1} \mathbf{X}$$

where $\mathbf{V}(\theta)$ is variance-covariance matrix of X_t at all lags

- ❖ In practice, use Cholesky decomposition to calculate $\det \mathbf{V}$ and $\mathbf{X}' \mathbf{V}^{-1} \mathbf{X}$

Application to likelihood-based estimation

❖ Given likelihood function $\mathcal{L}(\mathbf{X}; \theta)$ we can in turn do:

❖ Maximum likelihood estimation

$$\hat{\theta} = \arg \max \mathcal{L}(\mathbf{Y}, \theta)$$

❖ Bayesian estimation given prior $p(\theta)$: find posterior distribution

$$p(\theta | \mathbf{Y}) = \frac{\mathcal{L}(\mathbf{Y} | \theta) p(\theta)}{p(\mathbf{Y})} \propto \mathcal{L}(\mathbf{Y} | \theta) p(\theta)$$

❖ For instance, posterior mode just solves

$$\hat{\theta} = \arg \max p(\theta | \mathbf{Y})$$

[More complex: simulate from
posterior using MCMC]

Practical implementation: estimate HANK model!

- ❖ Notebook estimates HANK model with
 - ❖ 3 shocks (government spending, TFP, monetary policy)
 - ❖ 3 observables (output, inflation, interest rates)
- ❖ Look for
 - ❖ shock process parameters (assumed AR(1)'s: $3\sigma + 3\rho$)
 - ❖ coefficient on Taylor rule ϕ_π , fiscal rule ϕ_τ , slope of Phillips curve κ_w
- ❖ Done in less than a minute on a laptop!

Conclusion

- ❖ MIT shocks from SSJ are all you need for first-order solution!
 - ❖ Simulation, model moments, estimation
- ❖ Obtain same answer as state-space models linearized to first order
 - ❖ But speed gains can be significant
- ❖ More to come in next lecture, and then tomorrow: beyond certainty equivalence, MIT shocks are still all you need!